ACTIVITY 1 — PROGRESSIVE ELECTRICITY BILLING

1. Problem Statement

Develop a Python program to calculate the electricity bill based on the number of units consumed by a household. The program uses different rates for different consumption slabs and displays the total bill.

2. Objectives

1. To calculate the electricity bill based on units consumed.
2. To understand arithmetic operators and conditional statements.
3. To apply different rates to different consumption slabs.
4. To display the total electricity bill.

3. Assumptions

The following sample rates are used:

- First 100 units: ₹2 per unit.
- Next 100 units: ₹3 per unit.
- Above 200 units: ₹5 per unit.

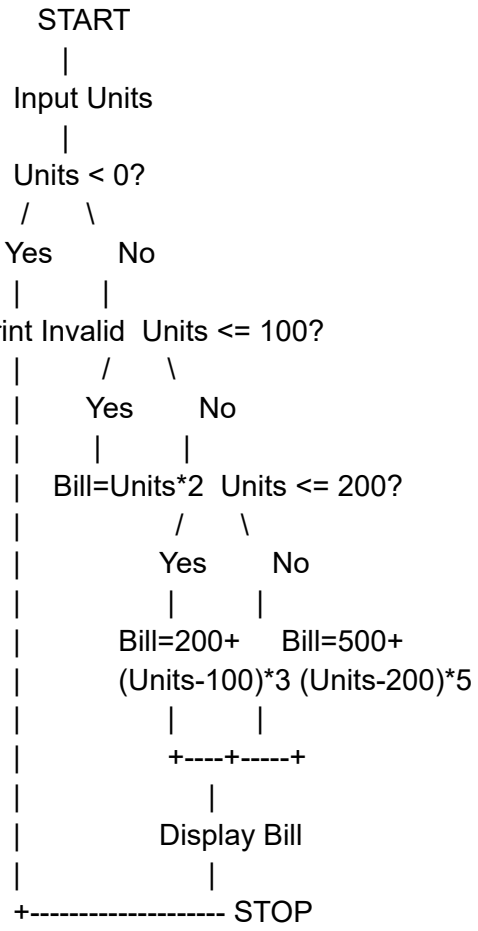
Other assumptions:

- Fixed charges and taxes are not included.
- Negative units are considered invalid.
- These are assumed rates for educational purposes only.

4. Algorithm

1. Start.
2. Read the electricity units consumed.
3. Check whether the units are negative.
4. If units are negative, display "Invalid units".
5. If units are at most 100, multiply units by 2.
6. If units are at most 200, calculate ₹200 plus ₹3 for each unit above 100.
7. Otherwise, calculate ₹500 plus ₹5 for each unit above 200.
8. Display the electricity bill.
9. Stop.

5. Flowchart



6. Pseudocode

```
START
INPUT units

IF units < 0
   PRINT "Invalid units"
ELSE IF units <= 100
   bill = units * 2
   PRINT bill
ELSE IF units <= 200
   bill = 200 + (units - 100) * 3
   PRINT bill
ELSE
   bill = 500 + (units - 200) * 5
   PRINT bill

STOP
```

7. Python Source Code

File name: `activity2.py`

```
units = int(input("Enter electricity units: "))

if units < 0:
    print("Invalid units")

elif units <= 100:
    bill = units * 2
    print("Electricity Bill: Rs.", bill)

elif units <= 200:
    bill = 200 + (units - 100) * 3
    print("Electricity Bill: Rs.", bill)

else:
    bill = 500 + (units - 200) * 5
    print("Electricity Bill: Rs.", bill)
```

8. Test Cases

Test Case	Input Units	Expected Output
1	50	Rs. 100
2	100	Rs. 200
3	150	Rs. 350
4	250	Rs. 750
5	-5	Invalid units

Sample Output

Enter electricity units: 150
Electricity Bill: Rs. 350

9. Output Screenshot and Conclusion

Output Screenshot: Execute the Python program in Python IDLE or VS Code. Enter a sample input and capture a screenshot showing the program execution and output. Save the screenshot as `activity2_output.png`.

Conclusion: The program calculates the electricity bill using progressive consumption slabs. It demonstrates the use of variables, input/output statements, arithmetic operators and conditional statements in Python.

Input code:

```
units = int(input("Enter electricity units: "))

if units < 0:
    print("Invalid units")

elif units <= 100:
    bill = units * 2
    print("Electricity Bill: Rs.", bill)

elif units <= 200:
    bill = 200 + (units - 100) * 3
    print("Electricity Bill: Rs.", bill)

else:
    bill = 500 + (units - 200) * 5
    print("Electricity Bill: Rs.", bill)
```

Output:

```
Enter electricity units: 50
Electricity Bill: Rs. 100

==========
Enter electricity units: 100
Electricity Bill: Rs. 200

==========
Enter electricity units: 150
Electricity Bill: Rs. 350

==========
Enter electricity units: 250
Electricity Bill: Rs. 750

==========
Enter electricity units: -5
Invalid units
```